EDUKRIPA

Science Mock Test - 3

Class Xth CBSE (2025 - 26) **Marks : 80**

Section - A

1. Photosynthesis is the process by which green plants prepare their own food. Which of the following are the essential raw materials for this process?
   (a) Oxygen, Water, and Sunlight (b) Carbon Dioxide, Oxygen, and Water
   (c) Sunlight, Chlorophyll, and Oxygen (d) Carbon Dioxide, Water, and Sunlight

2. Hormones regulate various functions in the human body. Which hormone is primarily responsible for the development of secondary sexual characteristics in males?
   (a) Oestrogen (b) Testosterone (c) Adrenaline (d) Insulin

3. Consider the following statements regarding the process of human digestion:
   (i) The digestion of starch begins in the mouth by the action of salivary amylase.
   (ii) Pepsin is the primary enzyme in the stomach that digests proteins in an acidic medium.
   (iii) Bile juice, produced by the liver, contains powerful enzymes for digesting fats.
   (iv) Trypsin, found in the small intestine, acts on proteins and breaks them down.
   Which of the above statements are correct?
   (a) (i) and (iii) only (b) (i), (ii), and (iv) (c) (ii) and (iv) only (d) All are correct.

4. The regulation of blood sugar levels is a classic example of a hormonal feedback mechanism. Which hormones are responsible for lowering and raising blood sugar levels, respectively?
   (a) Glucagon and Insulin (b) Insulin and Glucagon (c) Adrenaline and Thyroxine
   (d) Thyroxine and Adrenaline

5. The energy released during cellular respiration is captured to form a molecule that acts as the energy currency for the cell. What is this molecule?
   (a) ATP (b) DNA (c) RNA (d) ADP

6. Consider the following statements about cellular respiration:
   (i) Aerobic respiration occurs in the mitochondria and requires oxygen.
   (ii) Anaerobic respiration in human muscles produces ethanol and energy.
   (iii) Aerobic respiration yields a much larger amount of ATP than anaerobic respiration.
   (iv) The final step in anaerobic respiration is the breakdown of pyruvate into water and carbon dioxide.
   Which of the above statements are correct?
   (a) (i) and (iii) (b) (ii) and (iv) (c) (i), (ii), and (iii) (d) All are correct.

7. A deficiency of iodine in the diet can lead to a condition known as goitre, which is characterized by a swollen neck. This is caused by the enlargement of which gland?
   (a) Pituitary gland (b) Adrenal gland (c) Pancreas (d) Thyroid gland

8. Assertion (A) : Arteries are blood vessels with thick and elastic walls.
   Reason (R) : The blood emerges from the heart under high pressure and flows through the arteries
   (a) Both A and R are true, and R is the correct explanation of A.
   (b) Both A and R are true, and R is not the correct explanation of A.
   (c) A is true but R is false.
   (d) A is false but R is true.

9. Consider the following statements about the human excretory system:
   (i) The main organs of excretion are the pair of kidneys.
   (ii) The nephron is the basic filtration unit within the kidney.

(iii) The purpose of the system is to filter and remove nitrogenous wastes like urea from the blood.
(iv) All the fluid filtered into the nephron is excreted as urine without any reabsorption.
Which of the above statements are correct?
(a) (i) and (iv) only (b) (ii) and (iii) only (c) (i), (ii), and (iii) (d) All are correct.

10. In the human heart, the wall of the left ventricle is significantly thicker than the wall of the right ventricle. Provide a reason for this structural difference.

11. Attempt either option A or B.
A. Differentiate between autotrophic and heterotrophic modes of nutrition by giving one key difference and one example for each.

OR

B. What is lymph? State two major functions of the lymphatic system in the human body.

12. Attempt either option A and B.

A. Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds?

OR

B. What is transpiration, and how does it help in the upward movement of water in plants?

13. What is a synapse? Explain how a nerve impulse is transmitted from one neuron to the next across this gap.

14. Draw a diagram of a nerve cell and explain how nerve impulses are transmitted.

15. Draw a diagram showing the location of the major endocrine glands in the human body and label the following:
(i) Thyroid Gland
(ii) Adrenal Gland
(iii) Pituitary Gland

16. A person ate a piece of buttered toast.

A. The butter is rich in fats. Explain the role of Liver and its secretion in the digestion of fats.

B. The toast is rich in carbohydrates. Where does its digestion start and which enzyme from the Pancreas helps in its further digestion in the small intestine?

C. Explain the process of peristalsis and its role in Oesophagus.

Section - B

17. Analyze the following equations. Which statement correctly identifies the redox reaction and the values of 'p' and 'q' ?
Equation 1:
$pCuSO_4$ (aq) + Fe (s) $\longrightarrow$ $FeSO_4$ (aq) + Cu (s)
Equation 2:
$CaCO_3$ (s) $\xrightarrow{heat}$ CaO (s) + qCO_2 (g)
(a) Both are redox reactions; p = 1, q = 1
(b) Only Equation 1 is a redox reaction; p = 1, q = 1
(c) Only Equation 2 is a redox reaction; p = 1, q = 2
(d) Both are redox reactions; p = 2, q = 1

18. Most non-metal oxides are acidic in nature, but some are neutral. Which of the following is a neutral oxide?
(a) SO_2 (b) CO_2 (c) P_2O_5 (d) CO

19. A student performs experiments with four unknown metals A, B, C, and D. The observations are as follows:
1. Metal A displaces Metal B from its salt solution.
2. Metal B displaces Metal C from its salt solution.
3. Metal C displaces Metal D from its salt solution.
Based on these results, what is the correct order of decreasing reactivity?
(a) D C B A (b) A B C D (c) C B A D (d) B A D C

20. A curry stain on a white cloth is yellow in colour. When soap is scrubbed on it, the stain turns reddish-brown. What is the nature of soap and why does the colour change?
(a) Soap is acidic, and it reacts with the stain.
(b) Soap is neutral, and it cleans the stain.
(c) Soap is basic, and the turmeric in the curry acts as a natural indicator.
(d) Soap is acidic, and it bleaches the colour of the stain.

21. A student takes 10 mL of a hydrochloric acid solution (pH = 2) in a beaker and adds 90 mL of distilled water to it. What will be the effect on the pH of the final solution?
(a) The pH will decrease (b) The pH will increase (c) The pH will remain unchanged
(d) The pH will become 1

22. A student mixes a solution of dilute hydrochloric acid with a solution of sodium hydroxide. Which of the following correctly represents the products of this reaction?
(a) Sodium chloride and hydrogen gas (b) Sodium chloride, water, and chlorine gas
(c) Sodium oxide and water (d) Sodium chloride and water

23. A pale yellow powder, silver bromide, is kept in a china dish in the open. After some time, its colour changes to grey. This change is an example of a:
(a) Combination reaction (b) Photolytic decomposition reaction
(c) Displacement reaction (d) Precipitation reaction

24. Assertion (A) : Covalent compounds, like methane and ethanol, generally have low melting and boiling points.
Reason (R) : The intermolecular forces of attraction between the molecules of covalent compounds are very weak.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, and R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

25. Answer the following based on the physical properties of metals:
(a) Why are metals like gold and platinum, which are described as 'noble metals', exceptionally suitable for making jewellery? Give a reason based on their chemical property.
(b) Sodium is a metal, yet it is stored in kerosene and can be easily cut with a knife. What does this indicate about the physical hardness and chemical reactivity of sodium?

26. Attempt either option A or B.

A. A highly exothermic reaction between aluminium powder and iron(III) oxide is used for welding purposes on site.
(i) Give the common name of this reaction.
(ii) Write the balanced chemical equation for the reaction, indicating the physical states of the reactants and products.
(iii) Explain why the product of this reaction is useful for welding.

OR

B. Ionic compounds exhibit a unique set of physical properties due to the nature of the ionic bond.
(i) Explain why ionic compounds are generally hard, crystalline solids with high melting points.
(ii) Why do ionic compounds conduct electricity when dissolved in water or in the molten state, but not in the solid state?
(iii) Are ionic compounds typically soluble in an organic solvent like kerosene? Give a reason.

27. A student places a clean iron nail in a test tube containing a blue solution of copper sulphate. In another test tube, she places a copper wire in a freshly prepared pale green solution of ferrous sulphate.

(a) What changes would she observe in the first test tube containing the iron nail after 30 minutes?
(b) Write the balanced chemical equation for the chemical reaction that occurred in the first test tube.
(c) Would she observe any reaction in the second test tube? Justify your answer based on the reactivity series.

28. Attempt either the main question or the OR option.
A white chemical compound 'P' has the formula $CaSO_4.\frac{1}{2}H_2O$. It is prepared by carefully heating another calcium compound, Gypsum. This powder 'P' is widely used in hospitals and for making decorative items.
(a) Identify the compound 'P' and write its common name.
(b) Write the balanced chemical equation for the preparation of 'P' from Gypsum.
(c) Explain why 'P' must be stored in a moisture-proof container. Write the chemical equation for the change that occurs if it is not.

29. Bleaching powder is a yellowish-white solid used for various purposes, including disinfecting drinking water supplies and as a bleaching agent for cotton in the textile industry. It is produced by the action of a specific gas on dry slaked lime.
(a) What is the chemical name and formula for bleaching powder?
(b) Name the gas used in its preparation and write the balanced chemical equation for the reaction.
(c) What happens when bleaching powder is exposed to air for a long time? Explain why it loses its effectiveness.

Section - C

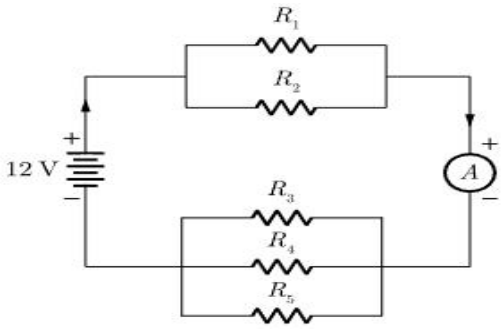
30. A student is studying about magnification produced by spherical mirrors and makes the following statements:
I. The magnification produced by a convex mirror is always positive and less than one.
II. The magnification produced by a concave mirror can be positive, negative, or one.
III. A magnification of -2 means the image is real, inverted, and twice the size of the object.
Choose from the following the correct option that lists the correct statements.
(a) I and II (b) I and III (c) I, II and III (d) II and III

31. During a laboratory experiment, a student observes that the path of light through a rectangular glass slab is bent twice. Choose the correct option that describes this phenomenon.
(a) Light undergoes two refractions: at the entry surface and at the exit surface.
(b) Light undergoes one reflection and one refraction.
(c) The bending of light is due to scattering by the glass molecules.
(d) The path of light is bent due to dispersion and reflection.

32. Assertion (A) : A convex lens always produces a real image of an object.
Reason (R) : A convex lens converges parallel rays of light to a single point, forming a real image.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, and R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

33. A. Explain what is meant by the principal focus of a concave mirror.
B. State the mathematical relationship between the radius of curvature and focal length of a sphericalmirror.
C. Draw a ray diagram for image formation when the object is placed at infinity.

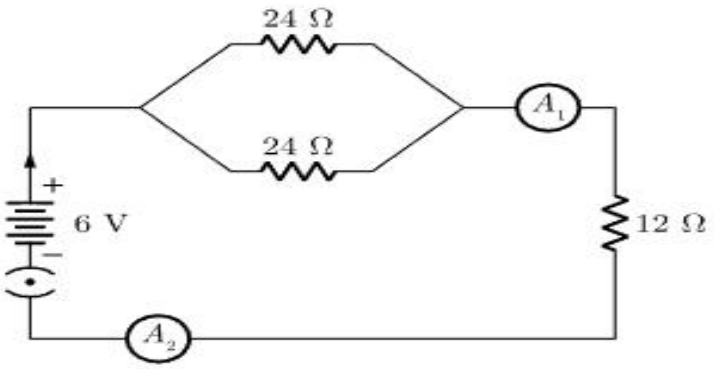
34. If in figure A, $R_1 = 10\Omega$, $R_2 = 40\Omega$, $R_3 = 30\Omega$, $R_4 = 20\Omega$, $R_5 = 60\Omega$ and a 12 V battery are connected to the arrangement, calculate :



(a) Total resistance in the circuit and
(b) Total current flowing in the circuit.

35. A wire is cut into three equal parts and then connected in parallel. How will its :
(a) Resistance
(b) Resistivity get effected ?

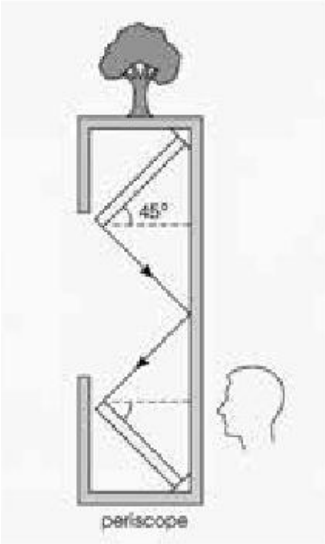
36. (i) How will you infer with the help of an experiment that the same current flows through every part of the circuit containing three resistors R1 , R2 and R3 in series connected to a battery of V volts?
(ii) Study the following circuit and find out:
(a) Current in 12Ω resistor.
(b) Difference in the readings of A_1 and A_2



37. Is there a relationship between the radius of curvature R , and focal length f , of a spherical mirror ? For spherical mirrors of small apertures, the radius of curvature is found to be equal to twice the focal length. We put this as R = 2f . This implies that the principal focus of a spherical mirror lies midway between the pole and centre of curvature.
(i) Write relation between radius of curvature and focal length.
(ii) For which type of mirrors above relation is verified?
(iii) What should be size of the aperture ?
R
(iv) Where is the principle focus of a spherical mirror lies?

38. Two electric lamps rated 100 W, 220 V and 25 W, 220 V are connected in parallel. Calculate the total electric current in the circuit.

39. The diagram shows a simple periscope used in submarines. A periscope is an optical device that allows sailors to see objects above the water surface while remaining underwater. It consists of two plane mirrors fixed at 45° to the vertical, placed parallel to each other inside a tube. Light from an object on the surface strikes the upper mirror, reflects downward to the lower mirror, and then reflects again into the observer's eye. The principle of reflection of light is used here. This device is also applied in some tanks and other defense instruments for observation.



A. Which law of reflection is followed by the plane mirrors in the periscope?

B. State two uses of periscopes apart from submarines.

Attempt either subpart C or D.

C. An object is placed 2.0 m in front of the upper mirror of a periscope. Find the distance of the final image from the observer's eye inside the periscope, if the two mirrors are separated by 50 cm vertically.

OR

D. A ray of light strikes the upper mirror at an angle of incidence 45°. Draw the path of the ray through the periscope, marking all angles clearly. Also, state the final direction of the ray emerging towards the observer.